main.c

Share Run

```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
struct Node* head = NULL;
void push(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct
        Node));
    newNode->data = value;
    newNode->next = head;
    head = newNode;
    printf("Successfully pushed %d onto the stack!\n", value);
}
int main() {
    push(10);
    push(20);
    push(30);

    return 0;
}
```

Output

Clear

```
Successfully pushed 10 onto the stack!
Successfully pushed 20 onto the stack!
Successfully pushed 30 onto the stack!


=== Code Execution Successful ===
```

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```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
struct Node* head = NULL;
void push(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->next = head;
    head = newNode;
    printf("Pushed: %d\n", value);
}
void pop() {
    if (head == NULL) {
        printf("Stack is empty, nothing to pop!\n");
        return;
    }
    struct Node* temp = head;
    head = head->next;
    printf("Popped value: %d\n", temp->data);
    free(temp);
}
int main() {
    pop();
    push(10);
    push(20);
    push(30);
    pop();
    pop();
    return 0;
}
```

Output

Clear

```
Stack is empty, nothing to pop!
Pushed: 10
Pushed: 20
Pushed: 30
Popped value: 30
Popped value: 20


=== Code Execution Successful ===
```

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Share Run

```c
    printf("Pushed: %d\n", value);
}
void pop() {
    if (head == NULL) {
        printf("Stack Underflow! Cannot pop.\n");
        return;
    }
    struct Node* temp = head;
    head = head->next;
    printf("Popped: %d\n", temp->data);
    free(temp);
}
void peek() {
    if (head == NULL) {
        printf("Stack is empty!\n");
    } else {
        printf("Current Top: %d\n", head->data);
    }
}
int isEmpty() {
    return (head == NULL);
}
int main() {
    printf("Is stack empty? %s\n", isEmpty() ? "Yes" : "No");
    push(100);
    push(200);
    peek();
    pop();
    peek();
    printf("Is stack empty now? %s\n", isEmpty() ? "Yes" : "No");

    return 0;
}
```

Output

Clear

```
Is stack empty? Yes
Pushed: 100
Pushed: 200
Current Top: 200
Popped: 200
Current Top: 100
Is stack empty now? No


=== Code Execution Successful ===
```

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```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
struct Node* head = NULL;
void push(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->next = head;
    head = newNode;
    printf("Pushed: %d\n", value);
}
void pop() {
    if (head == NULL) {
        printf("Stack Underflow! Cannot pop.\n");
        return;
    }
    struct Node* temp = head;
    head = head->next;
    printf("Popped: %d\n", temp->data);
    free(temp);
}
void peek() {
    if (head == NULL) {
        printf("Stack is empty!\n");
    } else {
        printf("Current Top: %d\n", head->data);
    }
}
int isEmpty() {
    return (head == NULL);
```

Output

Clear

```
Is stack empty? Yes
Pushed: 100
Pushed: 200
Current Top: 200
Popped: 200
Current Top: 100
Is stack empty now? No


=== Code Execution Successful ===
```

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